

JAVA PRACTICAL RECORD

Prime Palindrome

Experiment No.	02
Date	29/08/2026
Program	PrimePalindrome
Concept	Prime Number and Palindrome Checking

To determine whether a given number is both a prime number and a palindrome using Java.

Experiment 02 — Prime Palindrome

1. Objective / Aim

To write a Java program that accepts a number using Scanner and checks whether the number is both prime and palindrome.

2. System Requirements

Hardware: Standard PC/Laptop with 4 GB/8 GB RAM and a Core i3/i5 processor or equivalent.

Software: Windows/Linux/macOS, JDK 17 or JDK 21, and a Java editor or IDE such as VS Code, Eclipse, IntelliJ IDEA, or NetBeans.

3. Algorithm / Logic Description

1. Start the program.
2. Create a Scanner object and read an integer from the user.
3. Check whether the number is prime. If the number is less than or equal to 1, mark it as non-prime. Otherwise, test divisibility from 2 to $n/2$.
4. Store the original number and reverse its digits using a while loop.
5. Compare the original number with its reverse to determine whether it is a palindrome.
6. If both prime and palindrome conditions are true, display that the number is a Prime Palindrome.
7. Otherwise, display that the number is not a Prime Palindrome.
8. Close the Scanner and stop the program.

4. UML Class Diagram

PrimePalindrome
- sc : Scanner
- n : int
- prime : boolean
- original : int
- reverse : int
- digit : int
+ main(args : String[]) : void

5. Source Code

```
import java.util.Scanner;

public class PrimePalindrome {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter a number: ");
        int n = sc.nextInt();

        // Check Prime
        boolean prime = true;

        if (n <= 1) {
```

```

        prime = false;
    } else {
        for (int i = 2; i <= n / 2; i++) {
            if (n % i == 0) {
                prime = false;
                break;
            }
        }
    }

    // Check Palindrome
    int original = n;
    int reverse = 0;

    while (n > 0) {
        int digit = n % 10;
        reverse = reverse * 10 + digit;
        n = n / 10;
    }

    if (prime && original == reverse) {
        System.out.println(original + " is a Prime Palindrome.");
    } else {
        System.out.println(original + " is not a Prime Palindrome.");
    }

    sc.close();
}
}

```

6. Compilation & Execution Commands

```

javac PrimePalindrome.java
java PrimePalindrome

```

7. Output Screen

```

Enter a number: 131
131 is a Prime Palindrome.

```

Another test:

```

Enter a number: 121
121 is not a Prime Palindrome.

```

8. Result

The program successfully checks whether the entered number is both prime and palindrome using Java.